Experiment 5

PL/SQL with Exception Handling and Transaction Control
Student table | Oracle Database 21c XE

1. Source Code

```
-- ------------------------------------------------------------
-- Create the student table
-- ------------------------------------------------------------

-- 5-1. Create the student table and insert 15 rows
DROP TABLE student;
CREATE TABLE student (
  student_id   NUMBER PRIMARY KEY,
  student_name VARCHAR2(30),
  course       VARCHAR2(20),
  marks        NUMBER(3)
);

INSERT ALL
  INTO student VALUES (1,'Ravi','CSE',78)
  INTO student VALUES (2,'Sita','ECE',65)
  INTO student VALUES (3,'Arun','ME',55)
  INTO student VALUES (4,'Latha','CE',82)
  INTO student VALUES (5,'Kiran','EEE',48)
  INTO student VALUES (6,'Meena','CSE',91)
  INTO student VALUES (7,'Varun','CSM',60)
  INTO student VALUES (8,'Divya','CSD',73)
  INTO student VALUES (9,'Hari','ECE',39)
  INTO student VALUES (10,'Anu','CSE',67)
  INTO student VALUES (11,'Teja','ME',58)
  INTO student VALUES (12,'Neha','CE',85)
  INTO student VALUES (13,'Manoj','EEE',52)
  INTO student VALUES (14,'Lasya','CSM',70)
  INTO student VALUES (15,'Charan','CSD',44)
SELECT * FROM dual;

COMMIT;
SELECT * FROM student;

-- ------------------------------------------------------------
-- 5a. Cursor and user-defined exception (First Class students)
-- ------------------------------------------------------------

-- 5a-1. Display First Class students (marks >= 60)
SET SERVEROUTPUT ON
DECLARE
  CURSOR c_first IS
    SELECT student_id, student_name, marks
    FROM student
    WHERE marks >= 60;
  v_found BOOLEAN := FALSE;
  no_first_class EXCEPTION;
BEGIN
  FOR r IN c_first LOOP
    v_found := TRUE;
    DBMS_OUTPUT.PUT_LINE(r.student_id || ' ' ||
                         r.student_name || ' ' || r.marks);
  END LOOP;

  IF NOT v_found THEN
    RAISE no_first_class;
  END IF;
EXCEPTION
  WHEN no_first_class THEN
    DBMS_OUTPUT.PUT_LINE('No First Class Students Found.');
  WHEN OTHERS THEN
    DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
END;
/

-- 5a-2. No student satisfies the condition (change the condition to marks >= 95 and run again)
-- same program with:  WHERE marks >= 95;
-- ------------------------------------------------------------
```

```
-- 5b. COMMIT, ROLLBACK and SAVEPOINT
-- ------------------------------------------------------------

-- 5b-1. Insert records with SAVEPOINT, ROLLBACK TO SAVEPOINT and COMMIT
SET SERVEROUTPUT ON
BEGIN
  INSERT INTO student VALUES (16,'Kavya','CSE',72);
  INSERT INTO student VALUES (17,'Rahul','ECE',65);
  SAVEPOINT sp1;
  INSERT INTO student VALUES (18,'Sneha','ME',58);
  DBMS_OUTPUT.PUT_LINE('All three records inserted.');

  ROLLBACK TO sp1;
  DBMS_OUTPUT.PUT_LINE('Rollback to savepoint SP1 completed.');

  COMMIT;
  DBMS_OUTPUT.PUT_LINE('Transaction committed successfully.');
EXCEPTION
  WHEN OTHERS THEN
    ROLLBACK;
    DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
END;
/

SELECT * FROM student;
```

2. Execution and Output

Output shown is sample output of the programs above (Oracle Database 21c XE, SQL*Plus).

Create the student table

```
[5-1]
Table dropped.

Table created.

15 rows created.

Commit complete.

STUDENT_ID STUDENT_NAME COURSE MARKS
---------- ------------ ------ -----
         1 Ravi         CSE       78
         2 Sita         ECE       65
         3 Arun         ME        55
         4 Latha        CE        82
         5 Kiran        EEE       48
         6 Meena        CSE       91
         7 Varun        CSM       60
         8 Divya        CSD       73
         9 Hari         ECE       39
        10 Anu          CSE       67
        11 Teja         ME        58
        12 Neha         CE        85
        13 Manoj        EEE       52
        14 Lasya        CSM       70
        15 Charan       CSD       44

15 rows selected.
```

5a. Cursor and user-defined exception (First Class students)

```
[5a-1]
1 Ravi 78
2 Sita 65
4 Latha 82
6 Meena 91
7 Varun 60
8 Divya 73
10 Anu 67
12 Neha 85
14 Lasya 70

PL/SQL procedure successfully completed.

[5a-2]
No First Class Students Found.

PL/SQL procedure successfully completed.
```

5b. COMMIT, ROLLBACK and SAVEPOINT

```
[5b-1]
All three records inserted.
Rollback to savepoint SP1 completed.
Transaction committed successfully.

PL/SQL procedure successfully completed.

STUDENT_ID STUDENT_NAME COURSE MARKS
---------- ------------ ------ -----
         1 Ravi         CSE       78
         2 Sita         ECE       65
         3 Arun         ME        55
         4 Latha        CE        82
         5 Kiran        EEE       48
         6 Meena        CSE       91
         7 Varun        CSM       60
         8 Divya        CSD       73
         9 Hari         ECE       39
        10 Anu          CSE       67
        11 Teja         ME        58
        12 Neha         CE        85
        13 Manoj        EEE       52
        14 Lasya        CSM       70
```

```
        15 Charan       CSD        44
        16 Kavya        CSE        72
        17 Rahul        ECE        65

17 rows selected.
```